

CHEMICAL BONDING & MOLECULAR STRUCTURE

A Comprehensive, Concise Revision Formula & Concept Reference Sheet

1. IONIC (ELECTROVALENT) BONDING

Ionic Bond Formation: Formed when there is a **large difference in electronegativity** between combining elements (typically a reactive metal and a non-metal).

Lattice Enthalpy ($\Delta H_{\text{lattice}}$): Defined as the energy required to **completely separate one mole of a solid ionic compound** into its constituent gaseous ions.

Lattice Enthalpy Key Insight: Higher ionic charges and smaller ionic sizes lead to significantly higher lattice enthalpies. Example: Among NaCl, MgO, KCl, and CaCl_2 , **MgO has the highest lattice enthalpy** due to the +2 and -2 charges of Mg^{2+} and O^{2-} and their relatively small sizes.

Fajan's Rules (Covalent Character in Ionic Compounds):

No ionic bond is 100% ionic; covalent character is introduced by polarization. Polarizing power of the cation and polarizability of the anion govern this character:

- **Covalent character is favoured by:** (1) **Small size** of the cation, (2) **Large size** of the anion, and (3) **High charge** on both cation and anion.
- **Pseudo-Noble-Gas Configuration:** Cations with an 18-electron valence shell configuration in their outermost shell (such as Cu^+ , d^{10} configuration) possess **greater polarizing power** than inert-gas configuration cations of similar size and charge, thereby inducing **greater covalent character** in the compound.

2. COVALENT BONDING & LEWIS STRUCTURES

Octet Rule: Atoms tend to gain, lose, or share electrons so as to attain **eight electrons in their valence shell**, achieving a stable electronic configuration resembling the nearest noble gas.

Exceptions to the Octet Rule:

- **Incomplete Octet (Electron-Deficient Central Atoms):** Central atoms having fewer than 8 valence electrons. Examples include BF_3 and BeCl_2 .
- **Expanded Octet (Hypervalent Central Atoms):** Central atoms in elements of the 3rd period or beyond that can accommodate more than 8 valence electrons in their d-orbitals. Examples include SF_6 and PCl_5 .

Formal Charge Calculation:

The formal charge (Q) on any atom in a Lewis structure is given by the expression:

Formal Charge (Q) Formula:

$$Q = [V] - [L] - \frac{1}{2}[B]$$

Where: **[V]** = Total number of valence electrons in the free atom; **[L]** = Total number of non-bonding (lone pair) electrons; and **[B]** = Total number of bonding (shared) electrons.

Application Example (Carbonate Ion, CO_3^{2-}): In its resonance structures, the overall charge of -2 is shared equally among three oxygen atoms, yielding an **average formal charge of -2/3** on each oxygen atom.

3. VSEPR THEORY & MOLECULAR SHAPES

Valence Shell Electron Pair Repulsion (VSEPR) Theory: Explains molecular geometries based on minimizing the electrostatic repulsion between electron pairs in the valence shell of the central atom.

Order of Repulsive Strength between electron pairs:

Lone Pair - Lone Pair (lp-lp) > Lone Pair - Bond Pair (lp-bp) > Bond Pair - Bond Pair (bp-bp)

Bond Angle Compression: Due to the higher repulsive strength of lone pairs, bond angles compress. For instance, the **H-O-H bond angle in water (H_2O)** is compressed below the ideal tetrahedral angle of 109.5° because the **lp-lp repulsion exceeds bp-bp repulsion**, squeezing the O-H bonds closer together.

Summary of Molecular Shapes:

Bond Pairs (BP)	Lone Pairs (LP)	Molecular Shape	Key Example
3	0	Trigonal Planar	BF ₃
4	0	Tetrahedral	CH ₄
4	1	See-saw	SF ₄
3	2	T-shaped	ClF ₃
4	2	Square Planar	XeF ₄
5	1	Square Pyramidal	IF ₅

4. HYBRIDISATION

Concept: Intermixing of atomic orbitals of slightly different energies of the same atom to form a new set of equivalent orbitals of identical shapes and energies (hybrid orbitals).

Hybridisation States Reference Table:

Hybridisation	S-Character	Central Atom Example	Representative Properties
sp	50.0%	BeCl ₂ , CO ₂	Linear geometry (180°). Carbon has the highest electronegativity when in this hybridisation state.
sp ²	33.3%	BF ₃	Trigonal planar geometry (120°).
sp ³	25.0%	CH ₄ , NH ₃ , H ₂ O	Tetrahedral arrangement (109.5° ideal). Real bond angles compress as lone pairs increase.
sp ³ d	20.0%	PCl ₅	Trigonal bipyramidal geometry.
sp ³ d ²	16.7%	SF ₆	Octahedral geometry.

Key Hybridisation Insights:

- **Same Hybridisation Pairs:** Molecules with identical steric numbers on the central atom share hybridisation. For instance, both **CO₂** and **BeCl₂** have a central atom in the **sp hybridisation** state.
- **Electronegativity vs. S-Character:** Electronegativity increases as the s-character of the hybrid orbital increases. Thus, the electronegativity of a carbon atom follows the order **sp³ < sp² < sp**, being highest in the **sp** state.

5. DIPOLE MOMENT & BOND POLARITY

Dipole Moment (μ): A vector quantity measuring bond polarity, defined as the product of charge separation (e) and distance (d): **μ = e × d**.

Percentage Ionic Character:

The ionic character of any covalent bond can be quantitatively estimated using the ratio of observed and theoretical dipole moments:

Percentage Ionic Character Formula:

$$\% \text{ Ionic Character} = \left[\frac{\mu_{\text{observed}}}{\mu_{\text{theoretical}}} \right] \times 100$$

Where **μ_{theoretical}** is the dipole moment calculated assuming 100% ionic character.

Critical Dipole Moment Cases & Comparisons:

- **Zero Dipole Moment with Polar Bonds:** Highly symmetrical molecules can possess polar bonds whose individual vector dipoles cancel out completely. Example: CO_2 has a net dipole moment of zero due to its linear structure, whereas SO_2 and H_2O have bent geometries and possess permanent, non-zero dipole moments.
- **Polar Molecule Pair:** A pair like NH_3 and H_2O consists of two molecules that both have a permanent, non-zero dipole moment.
- **NH_3 vs. NF_3 Dipole Anomaly:** The net dipole moment of NF_3 is lower than that of NH_3 . In NF_3 , the highly electronegative F atoms pull electron density away from nitrogen, meaning the N-F bond dipoles point in a direction that partly opposes the lone-pair dipole. In NH_3 , the N-H bond dipoles point towards the nitrogen atom, thereby reinforcing the lone-pair dipole.
- **AB_3 -type Geometrical Difference:** An AB_3 molecule can be trigonal planar with a zero dipole moment (e.g., BF_3) or pyramidal with a non-zero dipole moment (e.g., NH_3). This difference is primarily dictated by the presence or absence of a lone pair on the central atom.

6. VALENCE BOND THEORY, RESONANCE & BOND PARAMETERS

Resonance: When a single Lewis structure cannot describe a molecule, multiple contributing structures are used. Resonance structures differ **only in the arrangement of electrons** (bonding pairs and lone pairs) while the **positions of the atoms remain fixed**.

Resonance Hybrid: The actual physical structure of the molecule is a weighted average of all contributing resonance structures. It is **more stable than any single contributing structure** due to the delocalisation of electron density.

Key Bond Parameters & Proportionalities:

- **Bond Order:** Defined as the number of bonds between two atoms. It serves as an **inverse measure of bond length** (i.e., as Bond Order increases, Bond Length decreases).
- **Bond Length Order:** The carbon-to-carbon bond length increases in the order: $\text{C}\equiv\text{C} < \text{C}=\text{C} < \text{C}-\text{C}$.
- **Bond Enthalpy:** The energy required to break one mole of a bond. It is directly proportional to bond order; as bond order between two atoms increases, the **bond enthalpy increases**.

General Properties of Covalent Compounds:

1. They possess **comparatively low melting and boiling points** relative to ionic compounds.
2. They are **generally poor electrical conductors** in both solid and molten states.
3. Covalent bonds possess a **directional nature**, giving rise to distinct molecular geometries.

Note on Solubility: High solubility in polar solvents (like water) is **not** a general property of all covalent compounds without exception (many are hydrophobic/insoluble).

7. MOLECULAR ORBITAL THEORY (MOT)

Concept: Atomic orbitals combine to form molecular orbitals (bonding and antibonding) that are delocalised over the entire molecule. This theory explains the magnetic behaviour and fractional bond orders of diatomic species.

Bond Order (BO) Calculation:

The bond order of a homonuclear or heteronuclear diatomic species is calculated as:

Molecular Orbital Bond Order Formula:

$$\text{BO} = \frac{1}{2}(\text{N}_b - \text{N}_a)$$

Where: N_b = Number of electrons occupying bonding molecular orbitals; and N_a = Number of electrons occupying antibonding molecular orbitals.

Bond Orders of Representative Diatomic Species:

Species	Bond Order	Valence Shell Config & Magnetic Behaviour Key Insights
H_2^+	0.5	Single bonding electron in σ_{1s} molecular orbital.
He_2	0.0	Hypothetical species ; cannot exist because $N_b = N_a = 2$ (BO = 0).
Li_2	1.0	Contains a stable single bond in the σ_{2s} orbital (BO = 1).
O_2	2.0	Paramagnetic ; possesses two unpaired electrons in degenerate π^* antibonding orbitals .
O_2^+	2.5	Formed by removing 1 e^- from the π^* antibonding orbital; BO increases from 2 to 2.5 .
N_2	3.0	Highly stable homonuclear diatomic molecule with a triple bond.
N_2^-	2.5	Formed by adding 1 e^- to the π^* antibonding orbital; BO decreases from 3 to 2.5 .

8. HYDROGEN BONDING

Hydrogen Bonding: A special type of dipole-dipole attraction that occurs when a hydrogen atom is covalently bonded to a **highly electronegative atom** (such as F, O, or N) and is simultaneously attracted to a **lone pair on another highly electronegative atom** on a neighbouring or same molecule.

Types of Hydrogen Bonding:

- **Intramolecular Hydrogen Bonding:** Occurs within a single, isolated molecule. It is observed in **ortho-nitrophenol** (where the nitro and hydroxyl groups are adjacent and form a chelate ring).
- **Intermolecular Hydrogen Bonding:** Occurs between separate molecules of the same or different substances. This results in molecular association:
 - Boiling Point Anomaly: The unusually high boiling point of **water (H_2O)** compared to other hydrides of the same group (H_2S , H_2Se , H_2Te) is due to **extensive intermolecular hydrogen bonding** between separate water molecules, requiring high energy to break.

9. ELECTRON-DEFICIENT HYDRIDES & DIBORANE

Electron-Deficient Hydride: A compound that has fewer valence electrons than required to form conventional two-electron covalent bonds between all adjacent atoms. Examples include group 13 hydrides like **diborane (B_2H_6)**.

Diborane (B_2H_6) Bonding:

- Diborane contains two coplanar BH_2 groups connected by two bridging hydrogen atoms.
- The bridging B-H-B bonds are classic **three-centre two-electron (3c-2e) bonds**, commonly known as **banana bonds**.
- These banana bonds allow boron to attain structural stability despite the severe electron deficiency of the molecule.

10. ISOELECTRONIC SPECIES

Definition: Isoelectronic species are atoms, molecules, or ions that possess **the same number of electrons**. Due to this identical electron count, they often exhibit **comparable structures** and bonding states.

Example: N_2 (14 electrons) and CO (14 electrons) are isoelectronic and share a triple-bond configuration.